



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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PROJECT TITLE

A CAPSTONE PROJECT REPORT

***Development of an Inventory and Sales Management System Using Modern
Software Engineering Practices***

Course Code - CSA1014

&

Course Name - Software Engineering

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
ARTIFICIAL INTELLIGENCE & DATA SCIENCE**

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Under the Supervision

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “*Development of an Inventory and Sales Management System Using Modern Software Engineering Practices*” has been carried out by VIMALA K (192424276) and Jyothika S U (192524122) under the supervision of DR.M. Arivukarasi and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Data Science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, VIMALA K of the B.tech AI&DS , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “*Development of an Inventory and Sales Management System Using Modern Software Engineering Practices*” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Vimala K JYothika S U/ 192424276 192524112	Specific Responsibilities	Design Development Contribution	& Testing Analysis Contribution	& Report Contribution	Approx. Contribut ion (%)
Vimala K (19242 4276)	development, database integration, report preparation and PPT creation	Developed the backend modules, implemented database operations, integrated the frontend with the backend, and managed the overall system functionality	Performed backend testing, database testing, system integration testing, and analysed the results	Prepared the complete project report, documentation, and PowerPoint presentation	50%
JYothika S U (19252 4112)	Frontend design and user interface development	Designed and developed the user interface for the Inventory and Sales Management System using modern frontend technologie	Performed frontend testing, usability testing, and checked the responsiveness and functionality of the user interface	Contributed frontend screenshots, interface details, and testing information for the project documentation	50%
Total					100%

ABSTRACT

Inventory and sales management is an important activity for organizations and businesses that need to maintain accurate records of products, stock levels, customer information, and sales transactions. Traditional manual methods or poorly integrated systems can lead to errors in inventory tracking, duplicate records, inaccurate stock information, and difficulties in analysing sales performance. Therefore, there is a need for an efficient and centralized software solution that improves the management of inventory and sales operations.

This project, **“Development of an Inventory and Sales Management System Using Modern Software Engineering Practices,”** proposes a web-based application to simplify and automate inventory and sales management. The system provides features for product management, inventory monitoring, sales recording, customer management, user authentication, and report generation. The frontend provides a user-friendly interface that allows users to access and manage different system functions easily. The backend processes user requests, manages business logic, and performs database operations to ensure efficient and reliable data management.

The development methodology follows modern software engineering practices, including requirement analysis, system design, modular development, implementation, testing, and documentation. The application is designed using appropriate web technologies, while the backend and database are integrated to support secure data storage and efficient processing. Different testing procedures are performed to verify the functionality of individual modules, database operations, system integration, and user interface responsiveness.

The developed system successfully provides a centralized platform for managing inventory and sales activities. It helps reduce manual work, improves the accuracy of stock and sales records, and enables users to access important information efficiently. Testing results demonstrate that the major modules function correctly and integrate effectively to support the overall workflow.

In conclusion, the proposed Inventory and Sales Management System offers an efficient, reliable, and user-friendly solution for managing business inventory and sales operations. The system demonstrates how modern software engineering practices can be applied to develop a practical and scalable web-based application.

Keywords: Inventory Management, Sales Management, Web Application, Database Integration, Software Engineering, Automation.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
API	Application Programming Interface	2
DB	Database	3

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Inventory and sales management are essential functions for businesses that handle products and customer transactions. Organizations need an effective system to maintain accurate information about available products, stock quantities, sales records, customers, and overall business performance. Traditional methods based on manual registers, spreadsheets, or disconnected software systems can make these activities time-consuming and prone to errors.

With the growth of digital technologies, web-based applications provide an effective solution for managing business operations through a centralized platform. An Inventory and Sales Management System can automate routine activities such as adding products, updating stock, recording sales transactions, managing customer information, and generating reports.

The proposed project, **“Development of an Inventory and Sales Management System Using Modern Software Engineering Practices,”** focuses on designing and developing a web-based system for efficient inventory and sales management. The system consists of a user-friendly frontend interface and a backend that manages business logic, data processing, and database operations. Modern software engineering practices such as modular design, systematic development, testing, and documentation are used to improve the reliability and maintainability of the application.

The system aims to provide users with accurate and up-to-date information regarding inventory and sales activities. By integrating different modules into a centralized application, the proposed system can reduce manual effort and improve the efficiency of business operations.

1.2 Need for the Project

Effective inventory management is important for maintaining the correct quantity of products and avoiding problems such as overstocking, understocking, and incorrect stock records. Similarly, accurate sales management is necessary to maintain transaction records and analyse business performance.

The problem affects small and medium-sized businesses, shop owners, inventory managers, sales staff, and administrators who may depend on manual records or multiple disconnected systems. These methods can lead to data duplication, calculation errors, delayed updates, and difficulties in accessing information.

Existing manual and traditional inventory systems have several limitations:

- Stock records may not be updated immediately after sales.
- Manual data entry can result in human errors.
- Maintaining separate records for products, customers, and sales can create data inconsistency.
- Generating reports manually requires additional time and effort.
- Existing systems may lack a centralized and user-friendly interface.
- Data security and controlled access may be limited.

The engineering significance of this project lies in integrating multiple business functions into a single software system. The application must ensure accurate data processing while maintaining usability, database consistency, system performance, and reliability. Therefore, a properly designed and tested software solution is required to address these technical challenges.

1.3 Problem Statement

The Core Processing & Data Management Module is responsible for managing the central data operations of the Inventory and Sales Management System. The engineering problem is to process interconnected data related to products, inventory, customers, and sales while maintaining accuracy, consistency, and reliability.

The module must handle multiple interacting factors, including product availability, stock updates, sales transactions, database operations, data validation, and record retrieval. A sales transaction should automatically affect the corresponding inventory quantity, while invalid or incomplete data must be prevented from entering the database.

The technical complexity arises from maintaining data consistency when multiple operations depend on the same records. The system must balance fast data processing with reliable validation and accurate storage. It must also operate within practical constraints such as available computing resources, database performance, development time, and network connectivity.

Therefore, the problem is to develop a reliable core processing mechanism that efficiently manages data operations and maintains consistency between the inventory and sales modules.

1.4 Project Aim

The aim of the Core Processing & Data Management Module is to design and implement a reliable backend mechanism for processing business data and managing products, inventory, customers, and sales records accurately and efficiently.

1.5 Project Objectives

- To design a structured database for storing products, inventory, customers, and sales information.
- To develop backend functions for creating, retrieving, updating, and deleting system records.
- To implement data validation before processing and storing information.
- To automatically update inventory quantities based on sales transactions.
- To maintain consistency between related inventory and sales records.
- To implement efficient data retrieval and processing operations.
- To test database operations and backend functionality.
- To evaluate the reliability and accuracy of the module

1.6 Scope

The module includes:

- Product data management.
- Inventory and stock management.
- Customer data management.
- Sales transaction processing.
- Database integration.
- Data validation and error handling.
- Automatic stock updates after sales
- Backend processing and business logic.

Excluded

The following features are outside the scope of this module:

- Frontend user interface design.
- Payment gateway integration.
- Advanced machine learning-based sales prediction.
- Barcode and RFID hardware integration.
- Multi-branch synchronization.
- Mobile application development.

Assumptions

- The frontend sends valid requests to the backend.
- The database server is available during normal system operation.
- Users have appropriate authorization to access system functions.
- Product and inventory information is maintained in the database.

Boundary Conditions

The performance of the module depends on database connectivity, server resources, available memory, and the number of simultaneous requests. The module is primarily designed for small and medium-scale business operations.

1.7 Expected Engineering Outcome

The expected outcome is a functional **Core Processing & Data Management Module** that acts as the central processing component of the Inventory and Sales Management System.

The module is expected to provide:

- A structured and integrated data management system.
- Accurate processing of products, customers, inventory, and sales.
- Automatic inventory updates based on sales transactions.
- Data validation and error handling.
- Consistent information across interconnected modules.

The final output is a backend software component that supports the overall operation of the Inventory and Sales Management Sys

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant information was identified through research papers, technical documentation, software engineering resources, database management literature, and existing inventory management systems. The review focused on technologies and methods used for backend processing, database management, inventory control, sales processing, and data consistency.

Existing solutions were compared based on functionality, data accuracy, scalability, usability, integration capability, and limitations. Technical documentation for web development frameworks, database systems, and software engineering practices was also reviewed.

2.2 Review of Existing Work

Traditional Spreadsheet-Based Systems

Many small businesses use spreadsheets to manage inventory and sales information. These systems are easy to implement and have a low initial cost. However, manual data entry can lead to duplicate records, calculation errors, and inconsistent information. Spreadsheet-based approaches also provide limited automation when inventory quantities must be updated after sales.

Database Management Systems

Database management systems provide a structured approach for storing and retrieving business information. Relational databases use tables and relationships to organize products, customers, inventory, and transaction data. They improve data consistency and support efficient data retrieval. However, database systems require proper schema design and validation to avoid redundancy and integrity problems.

Web-Based Inventory Management Systems

Modern web-based systems integrate frontend interfaces with backend processing and databases. These systems allow users to access inventory and sales information through a centralized platform. Although such systems improve automation and accessibility, their performance depends on efficient backend design, secure data handling, and proper communication between different modules.

Commercial Solutions

Commercial inventory and sales management solutions provide advanced features such as analytics, reporting, barcode integration, and multi-user access. However, many commercial platforms may involve subscription costs, unnecessary features for small businesses, or limited customization.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Records	Simple and inexpensive	High probability of human errors and difficult data retrieval	Demonstrates the need for automation
Spreadsheet-Based System	Easy to use and low initial cost	Limited integration and data consistency	Shows the importance of centralized processing
Standalone Database System	Structured data storage and retrieval	May lack a user-friendly integrated interface	Supports the database design approach
Web-Based Management System	Centralized access and automation	Requires reliable backend and database integration	Closely related to the proposed module
Commercial Inventory Software	Provides advanced features and reporting	Can be costly and difficult to customize	Motivates the development of a customized solution

2.4 Research / Engineering Gap

Existing approaches often focus either on user interface functionality or basic data storage without adequately addressing the integration between inventory and sales processing. Manual and spreadsheet-based systems lack automatic synchronization between transactions and stock records.

Commercial systems provide comprehensive features but may be expensive or unnecessarily complex for small and medium-sized businesses. Therefore, there is a need for a modular and customized core processing system that provides accurate data management, automatic inventory updates, and reliable integration between interconnected business operations.

2.5 Proposed Contribution

The proposed Core Processing & Data Management Module contributes a centralized backend mechanism for managing interconnected business data. The module integrates product management, customer records, inventory operations, and sales transactions within a structured data processing workflow.

A major contribution is the automatic updating of inventory information when sales are recorded. Data validation and error handling are included to improve the reliability of operations. The modular design also allows the system to be integrated with the frontend and extended with additional features in the future.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Administrator	Manage products, inventory, customers, and sales records
Inventory Manager	View and update stock information
Sales Staff	Record sales transactions accurately
Business Owner	Access organized business data and reports
System Administrator	Maintain reliable data storage and system operations

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Product Management	Functional	Add, update, view, and delete product records successfully.
Inventory Management	Functional	Maintain accurate stock quantities.
Sales Processing	Functional	Store valid sales transactions correctly.
Automatic Stock Update	Functional	Update inventory after every successful sale.
Input Validation	Non-Functional	Process normal requests efficiently.
Reliability	Non-Functional	Store and retrieve records accurately.

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
Software Engineering Principles	Modular and maintainable design	Backend functions are separated logically.
Database Normalisation	Reduce unnecessary data redundancy	Information is organized into structured database tables.
OWASP Security Principles	Protect application and user data	Input validation and authentication are considered.
REST API Principles	Standard communication between	Frontend and backend exchange structured requests and responses.
Data Integrity Principles	Maintain accurate records	Controlled updates are used for inventory and sales data.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Product Processing	Create, update, retrieve, and delete products	Operations	High
Inventory Accuracy	Maintain correct stock quantities	Records	High
Sales Processing	Store valid transactions	Transactions	High
Data Validation	Reject invalid input	Percentage	High
Data Consistency	Synchronize related data	System Function	High
Database Access	Retrieve records efficiently	Response Operation	Medium
Maintainability	Use modular code structure	Design Property	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Spreadsheet-Based Data Management

This approach uses spreadsheet files to store products, inventory, and sales information. It has a low implementation cost but requires significant manual work and provides limited automation.

Alternative 2: Standalone Database Application

This solution uses a database to store and manage information through a dedicated application. It improves data organization but may have limited accessibility and integration capabilities.

Alternative 3: Web-Based Backend and Database System

This approach uses a centralized backend application connected to a database. It supports automated processing, integration with frontend modules, and structured data management

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	0.25	2	3	5
Cost	0.15	5	3	4
Safety	0.20	2	4	4
Sustainability	0.15	3	4	5
Reliability	0.25	2	4	5

3.5 Selected Design & Detailed Design

The web-based backend and database architecture was selected because it provides the best balance between performance, reliability, scalability, and accessibility. The design supports centralized data processing while allowing communication between the frontend interface and the database.

Core Processing Workflow

User Request → Backend Processing → Data Validation → Business Logic → Database Operation → Updated Data → Response to User

The Core Processing Module receives requests from the frontend and validates the input data. The backend then applies the required business logic before performing database operations. When a sales transaction is completed, the corresponding inventory information is updated. The processed result is returned to the frontend, ensuring communication between all major system components.

The frontend, backend, and database operate as interconnected subsystems. The frontend collects user input, the backend processes the request, and the database stores or retrieves the required information. The correct operation of each subsystem depends on reliable interfaces between them.

3.5 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Centralised Data base	Separate data files	Better consistency	database setup	Selected for reliable integrated data management
Web based backened	Local standalone processing	Centralized access	Requires server support	Selected for integration and accessibility
Automated stock update	Manual stock modification	Reduces errors	Requires additional business logi	Selected to maintain accurate inventory
Input validation	Direct data storage	Improves reliability	Adds processing steps	Selected to prevent invalid records
Modular Design	Single large application structure	Easier maintenance	Requires initial design effort	Selected for future scalability

3.6 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduced paper-based record management	Digital storage and processing	Web-based digital management selected
Ethics	Proper handling of customer and business information	Data privacy requirements	Controlled access and responsible data use considered
Health & Safety	Software-based project with minimal physical safety risk	User interaction through standard computer systems	Basic ergonomic and safe computing practices considered
Security	Protection of business and user data	Authentication and input validation requirements	Secure data management included in design

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Database failure	Medium	High	High	Regular backup and error handling	Medium
Invalid Data entry	High	High	High	Input validation	Low
Data inconsistency	Medium	High	High	Controlled database operations	Low
Unauthorised access	Medium	High	High	Authentication and access control	Medium
Sever downtime Medium	Low	High	Medium	Backup and recovery planning	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

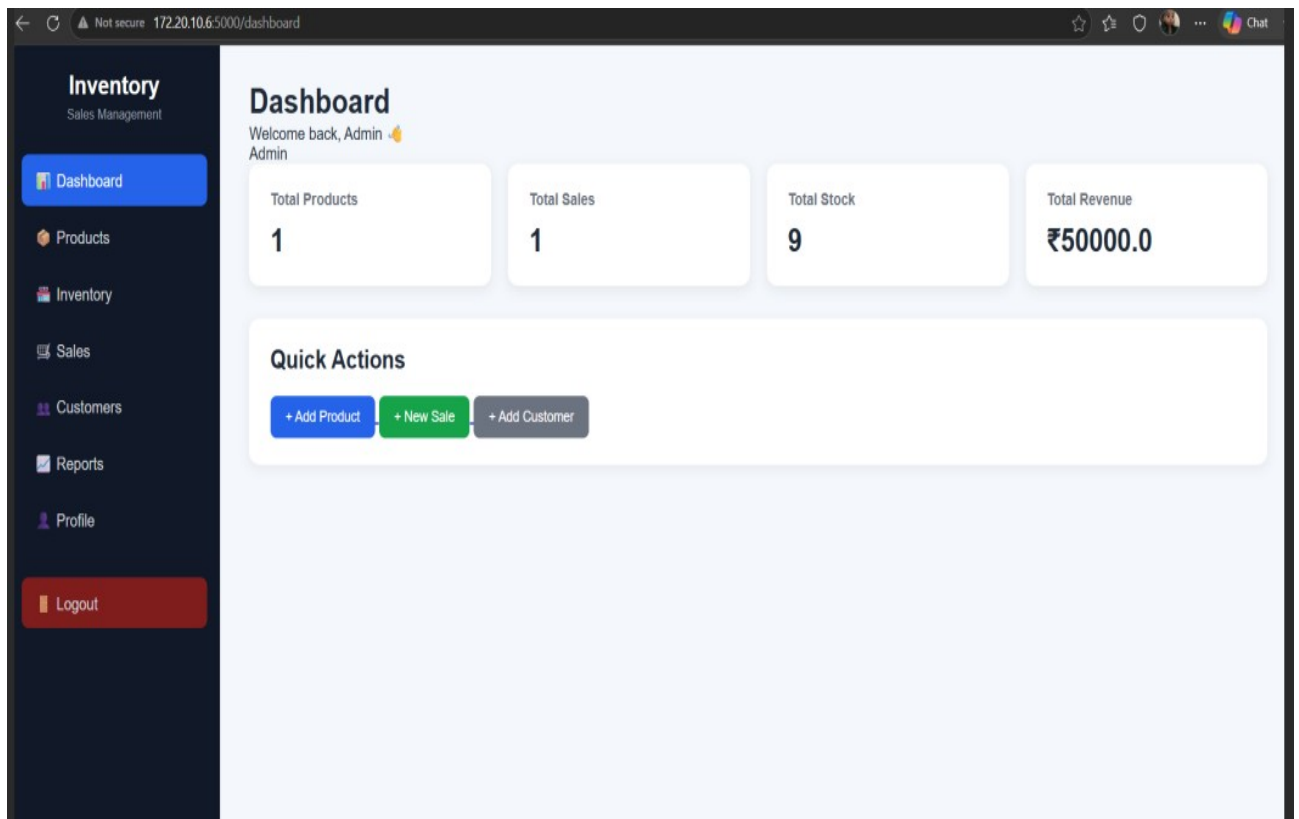
The Core Processing & Data Management Module was developed using a modular software engineering approach. The development process included requirement analysis, database design, backend implementation, integration, testing, and evaluation.

Each major function was implemented and verified separately before integration with the overall system. This approach improved maintainability and made it easier to identify and correct errors during development.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The module uses backend technologies to process business logic and manage communication between the frontend and database. Database operations are used to store and retrieve information related to products, customers, inventory, and sales.

Tool / Software / Equipment	Purpose	Stage Used
Python	Backend processing and business logic	Development
Flask	Web application and backend framework	Implementation
SQL/ Database	Data storage and management	Database Integration
HTML, CSS, JavaScript	Frontend communication	Integration
Visual Studio Code	Code development and debugging	Development
Git and GitHub	Version control	Project Management
Docker	Application containerization	Deployment and Testing



4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Add product	Functional testing	Product is successfully stored
Update Product	Functional testing	Updated details are displayed correctly
Record Sale	Functional testing	Sale record is stored correctly
Update Inventory	Integration testing	Stock quantity changes after a sale
Retrieve data	Database testing	Correct records are returned
Validate input	Negative testing	Invalid input is rejected

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Product Addition	Successful storage	Successful	Pass
Sales Recording	Accurate transaction record	Accurate	Pass
Sales Recording	Stock updated after sale	Update	Pass
Data Retrieval	Correct record returned	correct	Pass
Input Validation	Invalid data rejected	Reject	Pass
Backend Integration	Successful communication	Successfully	Pass

4.4 Results

The implementation of the Core Processing & Data Management Module successfully provides centralized processing for the Inventory and Sales Management System. Product, customer, inventory, and sales records can be processed and stored through the backend and database.

The testing results indicate that major database operations were successfully performed. Sales processing was integrated with inventory management so that stock information could be updated based on transaction activities. Data validation helped prevent incomplete or invalid information from being stored.

[Insert results tables / graphs / simulation outputs here.]

4.5 Data Analysis & Engineering Judgment

The test results demonstrate that integrating business logic with structured database operations improves the reliability of inventory and sales management. Automated processing reduces the dependency on manual record updates and improves data consistency.

The major engineering decision was to use a centralized backend and database architecture. Although this approach requires proper integration and database management, it provides greater reliability and maintainability than disconnected manual systems. Based on the testing results, the selected architecture is suitable for the intended small and medium-scale business environment.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Design structured data management	Database structure and module design	Fully Achieved
Develop backend processing	Functional backend operations	Fully Achieved
Implement sales processing	Integration test results	Fully Achieved
Update inventory automatically	Sales transaction functionality	Fully Achieved
Validate data	Input validation testing	Fully Achieved
Test module functionality	Functional and integration testing	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Centralized data processing.
- Structured database management.
- Automatic inventory updates.
- Improved data consistency.
- Modular backend architecture.
- Easier maintenance and future expansion.

Limitations:

- Designed primarily for small and medium-scale business operations.
- Advanced analytics are not included.
- Real-time multi-branch synchronization is not implemented.
- Performance may depend on database and server resources.
- Advanced hardware integration such as barcode scanning is outside the current scope.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
Vimala k	Backend development, database integration, core processing, documentation and PPT	Management Module and performed testing and documentation
Jyothika U S	Frontend development and user interface integration	Developed the frontend interface and supported integration with backend services
Item	Estimated Cost	Actual Cost
Software Development Tools	₹0	₹0
Visual Studio Code	₹0	₹0
Python ans Flask	₹0	₹0
Docker	₹0	₹0
github	₹0	₹0
Database Software	₹500	₹500
Internet and Computing Resources	₹500	₹500

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Core Processing & Data Management Module was developed to address the problem of managing interconnected inventory and sales information accurately and efficiently. The module provides centralized processing for products, inventory, customers, and sales records.

The proposed solution uses backend processing and database integration to perform data operations and maintain consistency between different business functions. Automatic inventory updates, input validation, and structured data management were implemented to improve reliability.

Functional and integration testing demonstrated that the major operations of the module were successfully implemented. Overall, the module contributes to the development of a reliable and maintainable Inventory and Sales Management System using modern software engineering practices.

5.2 Future Work

Future improvements may include:

- Integration of barcode and QR code scanning.
- Multi-branch inventory synchronization.
- Cloud database deployment.
- Advanced sales analytics and visualization.
- Machine learning-based sales forecasting.
- Automated low-stock notifications.
- Mobile application support.
- Role-based access control with advanced security features.
- Integration with online payment and billing systems.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Backend web application development.
- Database design and integration.

- CRUD operations.
- Business logic implementation.
- API communication.
- Software testing techniques.
- Version control and project documentation.

Engineering and Professional Skills Developed

- Problem-solving and debugging.
- Modular software design.
- Database management.
- System integration.
- Technical documentation.
- Testing and evaluation.
- Project planning and collaboration.

Individual Reflection – Vimala K

The most difficult engineering problem encountered during the development of this module was maintaining accurate synchronization between sales transactions and inventory records. This problem was addressed by implementing backend business logic that processes sales information and updates the related stock records systematically.

A key engineering decision was to use a centralized backend and database approach because it provides better data consistency and maintainability. This decision was supported by the limitations observed in manual and disconnected data management methods.

Through independent learning, I gained additional knowledge about backend development, database integration, testing, and modern software engineering practices. If the project were repeated, I would further improve the module by adding advanced security, cloud deployment, and real-time multi-user processing capabilities.

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APPENDICES

Appendix A – Detailed Calculations

No calculation-intensive derivations (e.g., signal, mechanical, or statistical proofs) were required for this software-focused module. This appendix is retained per the official format and left without content, as none is applicable.

Appendix B – Engineering Drawings / Schematics

See Figure 3.1 (Chapter 3) for the system architecture diagram. No physical engineering drawings apply to this software project.

Appendix C – Source Code / Code Repository Information

Per the official report format, complete source code is not reproduced in the main report body. The project's source code is maintained in the team's working repository. [Repository URL and access details to be supplied by the student/team — not fabricated here in the absence of confirmed repository information.]

Appendix D – Datasheets

Quantitative experimental data (accuracy logs, latency measurements, quiz-performance datasets) was not available at the time of this report, per Section 4.4–4.5. This section is to be populated once such data is collected.

Appendix E – Experimental / Raw Data

Not applicable — this course project did not involve human-subject research requiring formal institutional ethics approval; general ethical considerations relevant to system design are addressed in Section 3.7.

Appendix F – Ethics / Institutional Approval

[Meeting notes / progress-tracking records to be supplied by the student/team, if maintained, and attached here.]

Appendix G – Project Meeting / Progress Records

Not available at the time of this report — no formal user or industry feedback was collected for the adaptive-learning module. This is identified as a future-work item (Section 5.2).

Appendix H – User/Industry Feedback

Not applicable — no publications, patents, competition entries, or external product outcomes are associated with this course project at the time of this report.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable — no publications, patents, competition entries, or external product outcomes are associated with this course project at the time of this report.

Appendix J – Individual Contribution Evidence

Evidence supporting the individual contribution described in this report, submitted by Vimala K (192424276):

Operational system screenshot: Figure 4.1, adaptive assessment engine interface (Chapter 4).

Project presentation slides (team-submitted, referenced throughout this report as the project source material).

[Version-control commit history / individual task-tracking records to be supplied by the student, if maintained, and attached here.]

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)